

REPORT OF 17 YEARS OF EXPERIENCE IN MANAGEMENT OF ADULT AND PEDIATRIC INTESTINAL FAILURE PATIENTS IN A SINGLE REFERRAL CENTER IN ARGENTINA

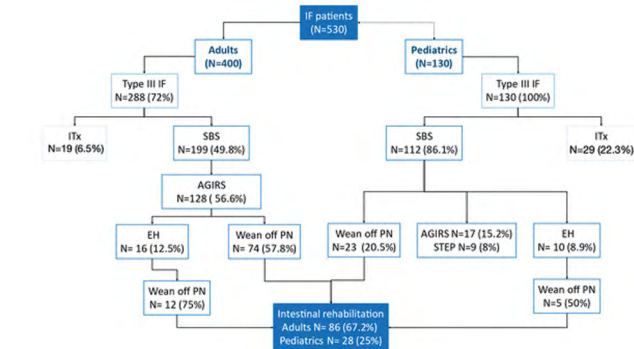
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Introduction: Parenteral nutrition (PN) is the first line treatment for intestinal failure (IF). In the last decades the development of intestinal rehabilitation strategies included both medical-nutritional treatment (drugs, feeding strategies, use of enterohormones [EH]) and surgical procedures (autologous gastrointestinal reconstruction surgery [AGIRS], lengthening procedures, transplantation [ITx]) with the aim of re-establishing intestinal autonomy. The aim of this study is to report the results of 17 years of experience in a specialized single center in Argentina.

Methods: From February 2006 to February 2023 a retrospective analysis of a prospective database of patients (pts) with IF was performed. Pts were divided into 2 groups (A adults and P pediatrics) reporting the main causes of IF. From those pts with short bowel syndrome (SBS) number of surgeries performed, EH treatment, PN independence and overall survival were assessed. ITx indications and outcomes were also reported.

Results: 530 pts with IF were included (Group A=400, P=130). In group A, IF was type I in 48 (12%), type II in 64 (16%) and type III in 288 (72%). SBS was the leading cause in 199 pts (49.8%) followed by intestinal fistula in 66 (16.5%), intestinal dysmotility in 65 (16.2%), mechanical obstruction in 46 (11.5%) and extensive small bowel mucosal disease in 24 (6%). From the SBS group, 128 pts (56.6%) received AGIRS. Mean post-surgical intestinal length was 144.3 ± 101.1cm; anatomy type was 1 in 13 (10.2%), 2 in 41 (32.0%) and 3 in 74 (57.8%). From those pts, 74 (57.8%) were weaned off PN. EH were used in 16 out of 128 pts (12.5%) and 12/16 (75%) achieved intestinal rehabilitation without severe adverse events, with a mean intestinal length of 47.5 ± 43.7cm. The overall survival at 10 years was 77%. Nineteen out of 288 pts (5.9%) received 20 ITx, and 89% achieved PN independence. ITx survival at 10-year was 41%. In group P IF was type III in all the cases and the leading causes were SBS in 112 (86.1%), intestinal dysmotility in 11 (8.5%) and extensive small bowel mucosal disease in 7 (5.4%). In the SBS group 17 pts received AGIRS (15.2%) and 9 lengthening procedures (8.0%). Mean intestinal length post initial resection was 28.7 ± 27cm. Twenty three out of 112 (20.5%) SBS pts were able to wean off PN and ten (8.9%) required EH treatment, achieving intestinal independence in 5 (50%) The overall survival was 69% at 10 years. Twenty-nine pts received 33 ITx, 69% achieved PN independence at 1 year, and 10 years survival was 44 %.

Conclusion: After 17 years of experience 530 pts with IF were treated in our program. The best results were obtained by a multidisciplinary approach, combining surgical, medical, and nutritional strategies. The introduction of EH improved the outcome of SBS pts, allowing to achieve intestinal sufficiency even in those with unfavourable intestinal length and anatomy types. ITx was required in only 10% of the cases.



IF: Intestinal failure; ITx: Intestinal transplant; SBS: Shortbowel syndrome; AGIRS: Autologous Gastrointestinal Reconstruction Surgery; STEP: serial transverse enteroplasty; PN: Parenteral Nutrition; EH: Enterohormones

Improved outcome in the past fifteen years of a pediatric intestinal failure Latin American cohort

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Introduction: Multidisciplinary (MD) Intestinal Rehabilitation Programs are crucial to obtain the best results in children with intestinal failure (IF). International multicenter reports describe better results on mortality, rehabilitation, and less need for transplantation in the last decade. Our aim: to analyze the outcomes of children with IF managed by a MD team at a reference center in Latin America.

Methods: Retrospective review of medical records from January 2008 to December 2022 of pediatric patients with IF managed by an intestinal rehabilitation MD team. Demographic data were collected, together with rehabilitation, mortality and transplantation rates. They were analyzed in the global population and also in two equal time periods, according to initiation of follow-up either before or after June 2015, the middle of the studied period. Early referral (before 6 months of diagnosis) and bowel lengthening procedures were recorded. The descriptive data was analyzed by chi2 test or Fisher's test for categorical variables and by t-test or Mann Whitney for continuous variables, depending on the distribution.

Results: 156 patients with IF were treated during the study period. Median age at first consultation was 0.43 yrs. (IQR 0.1–1.75), 95 males. The IF etiology was short bowel syndrome (SBS) in 127/156 (81.4%), motility disorder in 24 (15.4%) and congenital enteropathy in 5 patients (3.2%). Thirteen patients had type 1 SBS, 91 type 2, and 23 type 3. 86/156 were able to stop parenteral nutrition, with a rehabilitation rate of 55%, 2/91 received teduglutide. Enteral autonomy was achieved in a mean time of 1.03 years (SD ±1.25). Fourteen intestinal transplants were performed in 13 patients, of which 8 are still alive, 4 with functioning grafts. Cumulative mortality during these 15 years was 16% (25/156). The analysis between the two periods showed better trends in rehabilitation, with a statistically significant difference observed with lower mortality rates and less transplantations performed in patients from the most recent period.

There was also a tendency towards better rehabilitation rate, shorter time to reach it, and higher proportion of patients with early referral, although the difference was not statistically significant (Table 1).

Conclusion: Although the possibility of reaching intestinal rehabilitation continues to increase, the most important impact observed in this pediatric cohort with IF, was reduced mortality and less transplantation in the recent times supporting international reports.

Table 1	2008-2015 n: 85	2015-2022 n: 71	p
Mortality	20/85 (23.5%)	5/71 (7%)	.0076
Transplantation	12/85	1/71	.0065
Rehabilitation (R)	43/85 (50.6%)	43/71 (60.6%)	ns
Time to reach R (y (SD))	1.21 (±1.26)	0.85 (±1.24)	ns
Early Referral	47/85 (55.3%)	50/71 (70%)	ns
Surgery	28/85 (32.9%)	18/71 (25.3%)	ns